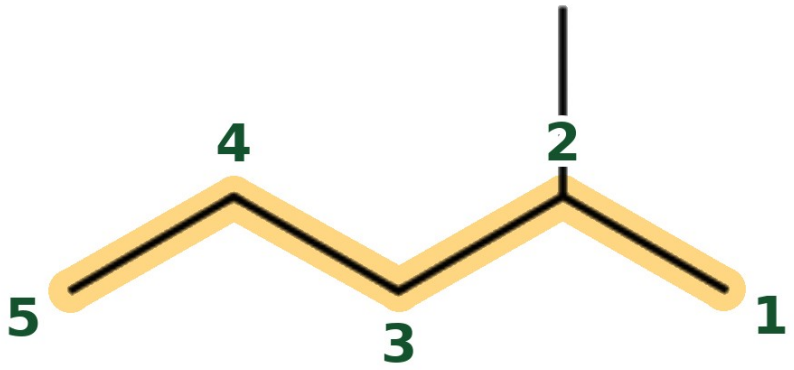
A01 FOUNDATION

Structure → Name

Molecular formula (counted): **C6H14**

(6 C, open chain: C_nH_{2n+2})

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 5 C → pentane (chain). *(Other paths of the same length: 5 C, using the top methyl branch instead of the rightmost carbon as a chain end.)*

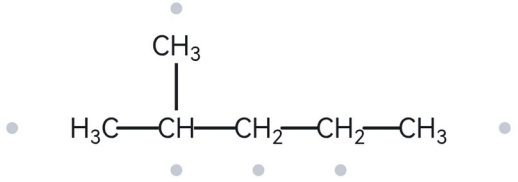
Numbering: right → left gives the substituent C2; left → right gives C4. Winner: right → left. *(2 is lower than 4.)*

Substituents (A→Z): methyl on C2.

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

2-methylpentane

4 · THE ALKANE CHECK (RETURNED NAME)



2-methylpentane

Tool: The Alkane

Returned name: 2-methylpentane

Verdict: [✓] Match [ ] Mismatch

[✓] longest chain/ring confirmed [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

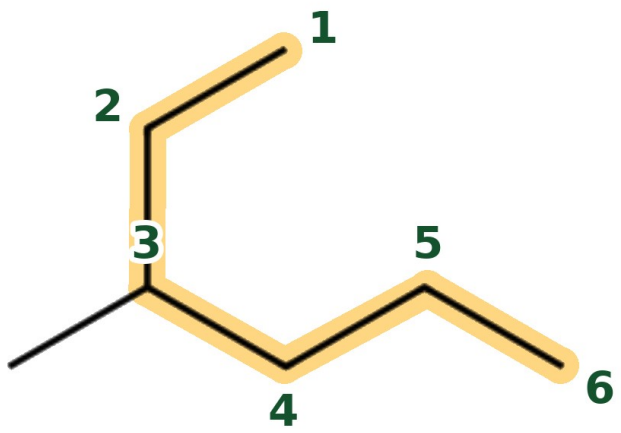
A02 FOUNDATION

Structure → Name

Molecular formula (counted): **C7H16**

(7 C, open chain: C_nH_{2n+2})

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 6 C → hexane (chain). *(The longest chain is not the horizontal one; it bends through the branch carbon. Other paths of the same length: none, the others give 4 C or 5 C.)*

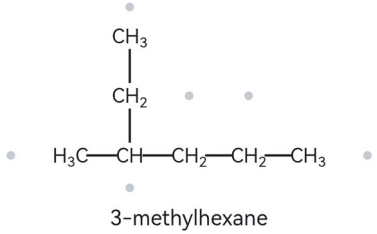
Numbering: from the top-right end down to the bottom-right end gives the branch C3; the other direction gives C4. Winner: C3.

Substituents (A→Z): methyl on C3.

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

3-methylhexane

4 · THE ALKANE CHECK (RETURNED NAME)



Tool: The Alkane

Returned name: 3-methylhexane

Verdict: [✓] Match [ ] Mismatch

[✓] longest chain/ring confirmed **[✓] lowest locants** **[✓] alphabetical order** **[✓] punctuation** **[✓] formula counted**

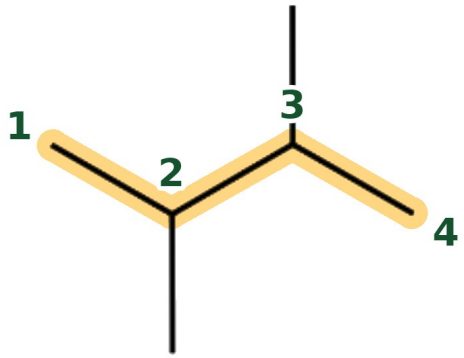
A03 FOUNDATION

Structure → Name

Molecular formula (counted): **C6H14**

(6 C, open chain: C_nH_{2n+2})

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 4 C → butane (chain). *(Other paths of the same length: 4 C, using either vertical methyl branch as a chain end.)*

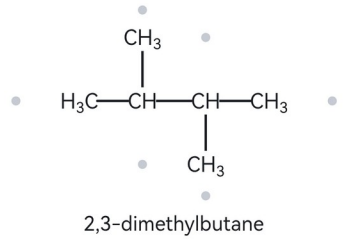
Numbering: left → right or right → left: the molecule is symmetrical, so the locants are 2 and 3 either way.

Substituents (A→Z): methyl on C2, methyl on C3. *(Identical groups, so they are combined with the prefix di-.)*

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

2,3-dimethylbutane

4 · THE ALKANE CHECK (RETURNED NAME)



Tool: The Alkane

Returned name: 2,3-dimethylbutane

Verdict: [✓] Match [ ] Mismatch

[✓] longest chain/ring confirmed **[✓] lowest locants** **[✓] alphabetical order** **[✓] punctuation** **[✓] formula counted**

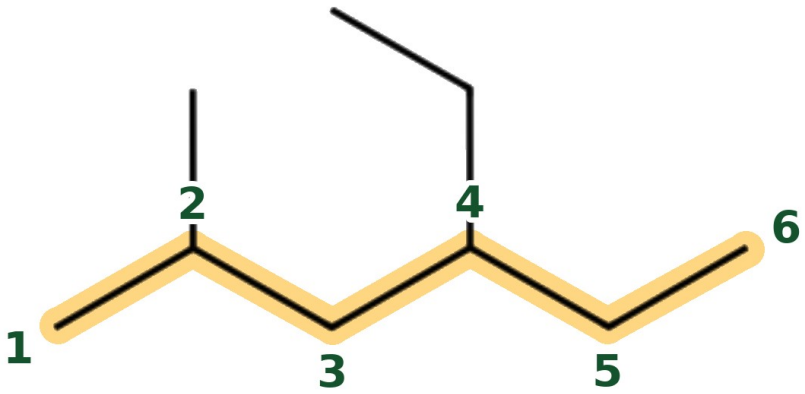
A04 DEVELOPING

Structure → Name

Molecular formula (counted): **C9H20**

(9 C, open chain: C_nH_{2n+2})

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 6 C → hexane (chain). *(Other paths of the same length: 6 C, tracing up through the ethyl branch instead of continuing to the rightmost tip.)*

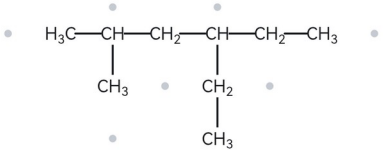
Numbering: left → right gives 2,4; right → left gives 3,5. Winner: left → right. *(First point of difference: 2 is lower than 3.)*

Substituents (A→Z): ethyl on C4, methyl on C2. *(Ethyl is cited first because "e" comes before "m" alphabetically.)*

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

4-ethyl-2-methylhexane

4 · THE ALKANE CHECK (RETURNED NAME)



4-ethyl-2-methylhexane

Tool: The Alkane

Returned name: 4-ethyl-2-methylhexane

Verdict: [✓] Match [ ] Mismatch

[✓] longest chain/ring confirmed **[✓] lowest locants** **[✓] alphabetical order** **[✓] punctuation** **[✓] formula counted**

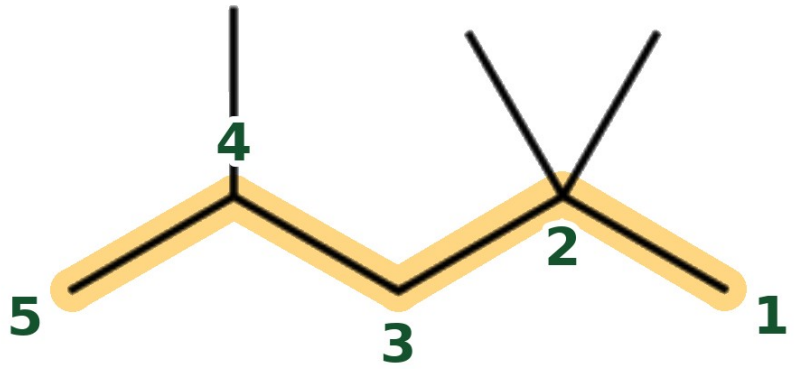
A05 DEVELOPING

Structure → Name

Molecular formula (counted): **C8H18**

(8 C, open chain: C_nH_{2n+2})

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 5 C → pentane (chain). *(Other paths of the same length: 5 C, using any of the branching terminal methyl groups as chain ends.)*

Numbering: right → left gives 2,2,4; left → right gives 2,4,4. Winner: right → left. *(First point of difference: 2 is lower than 4.)*

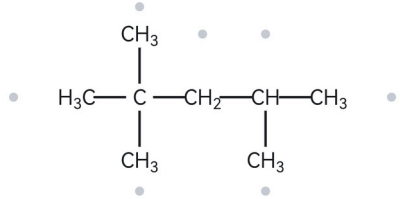
Substituents (A→Z): methyl on C2, methyl on C2, methyl on C4.

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

2,2,4-trimethylpentane

(commonly known as isooctane)

4 · THE ALKANE CHECK (RETURNED NAME)



2,2,4-trimethylpentane

Tool: The Alkane

Returned name: 2,2,4-trimethylpentane

Verdict: **[✓]** Match **[ ]** Mismatch

(Isooctane is the common name of this compound.)

[✓] longest chain/ring confirmed **[✓] lowest locants** **[✓] alphabetical order** **[✓] punctuation** **[✓] formula counted**

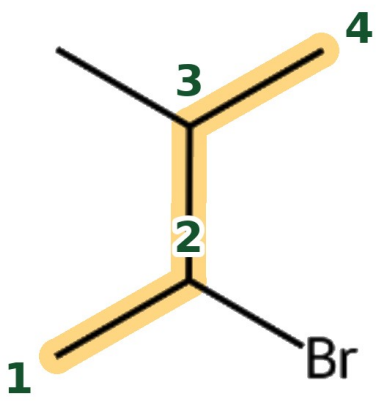
A06 DEVELOPING

Structure → Name

Molecular formula (counted): **C5H11Br**

(5 C chain C_5H_{12}, one H replaced by Br)

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 4 C → butane (chain). *(Other paths of the same length: 4 C, using either top methyl group as the chain start.)*

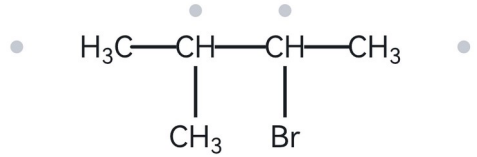
Numbering: bottom-left → top-left gives bromo C2, methyl C3; the other direction gives methyl C2, bromo C3. Both sets are 2,3 (a tie). *(Alphabetical tie-break: bromo comes before methyl, so bromo gets the lower number C2.)*

Substituents (A→Z): bromo on C2, methyl on C3.

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

2-bromo-3-methylbutane

4 · THE ALKANE CHECK (RETURNED NAME)



2-bromo-3-methylbutane

Tool: The Alkane

Returned name: 2-bromo-3-methylbutane

Verdict: [✓] Match [ ] Mismatch

[✓] longest chain/ring confirmed [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

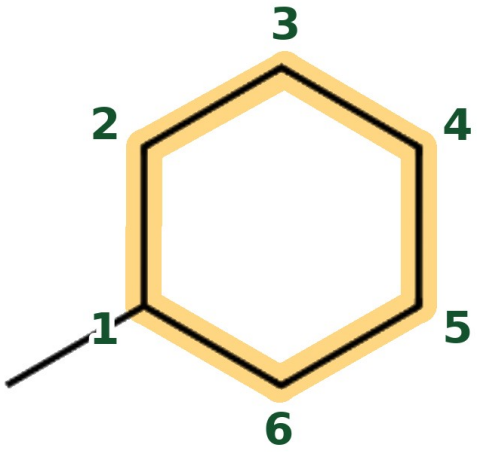
A07 DEVELOPING

Structure → Name

Molecular formula (counted): **C7H14**

(7 C, one ring: C_nH_{2n})

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 6 C → cyclohexane (ring). *(Other paths of the same length: none.)*

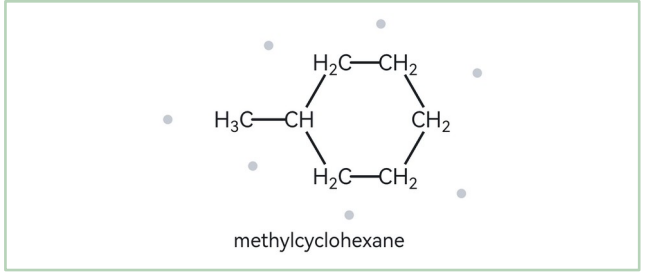
Numbering: no locant is needed. *(A single substituent on a ring is automatically assigned position C1.)*

Substituents (A→Z): methyl on C1.

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

methylcyclohexane

4 · THE ALKANE CHECK (RETURNED NAME)



Tool: The Alkane

Returned name: methylcyclohexane

Verdict: [✓] Match [ ] Mismatch

[✓] longest chain/ring confirmed [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

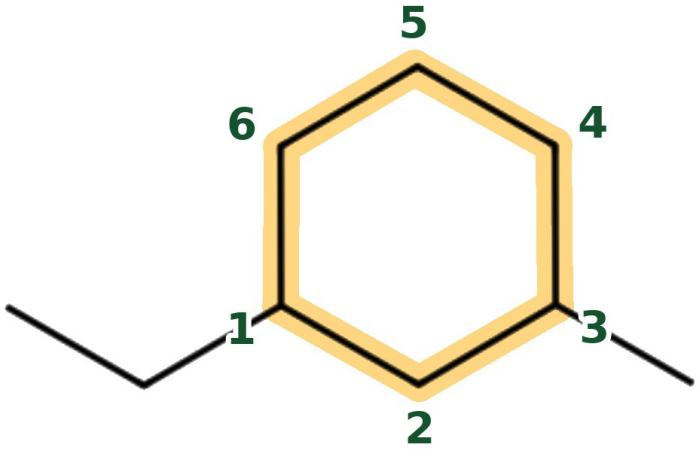
A08 CHALLENGE

Structure → Name

Molecular formula (counted): **C9H18**

(9 C, one ring: C_nH_{2n})

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 6 C → cyclohexane (ring). *(Other paths of the same length: none.)*

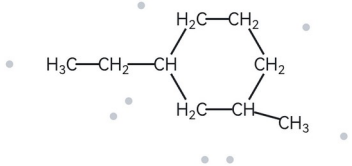
Numbering: counter-clockwise from the ethyl group to the methyl group gives {1,3}; the other way round also gives {1,3}. *(Alphabetical tie-break: ethyl comes before methyl, so ethyl gets C1.)*

Substituents (A→Z): ethyl on C1, methyl on C3.

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

1-ethyl-3-methylcyclohexane

4 · THE ALKANE CHECK (RETURNED NAME)



1-ethyl-3-methylcyclohexane

Tool: The Alkane

Returned name: 1-ethyl-3-methylcyclohexane

Verdict: [✓] Match [ ] Mismatch

[✓] longest chain/ring confirmed [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

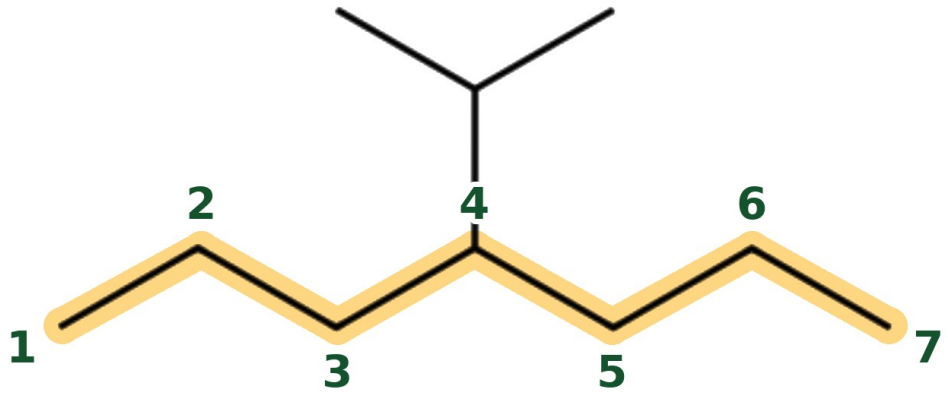
A09 CHALLENGE

Structure → Name

Molecular formula (counted): **C10H22**

(7 + 3 = 10 C, open chain: C_nH_{2n+2})

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 7 C → heptane (chain). *(Other paths: 6 C, tracing from an end into the branched substituent, so the 7 C chain is the longest.)*

Numbering: left → right or right → left: the chain is symmetrical, so the substituent is at C4 either way.

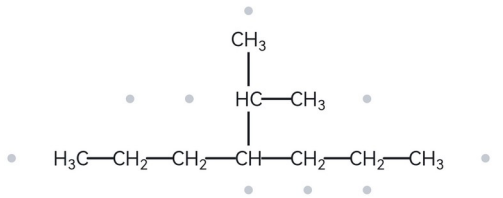
Substituents (A→Z): isopropyl on C4. *(Isopropyl = 1-methylethyl = propan-2-yl; all are the same group.)*

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

4-isopropylheptane

or 4-(1-methylethyl)heptane

4 · THE ALKANE CHECK (RETURNED NAME)



2-methyl-3-propylhexane

Tool: The Alkane

Returned name: 2-methyl-3-propylhexane

Verdict: [ ] Match [✓] Mismatch
Correct name: 4-isopropylheptane

Rule: parent = longest chain (7 C, not 6 C)

[✓] longest chain/ring confirmed **[✓] lowest locants** **[✓] alphabetical order** **[✓] punctuation** **[✓] formula counted**

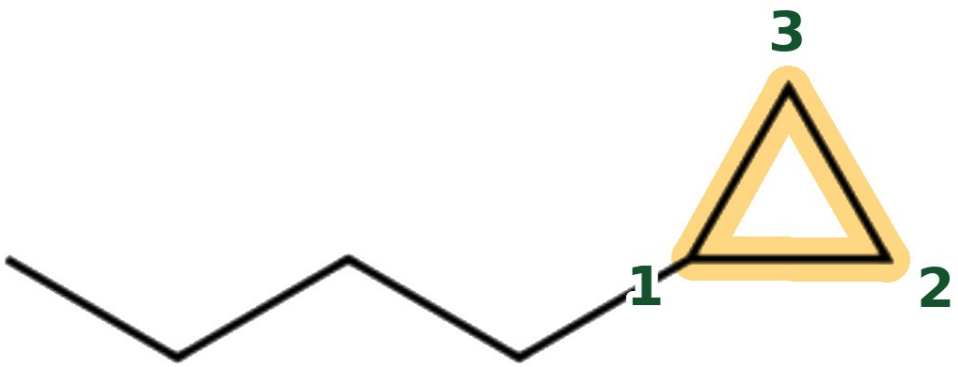
A10 CHALLENGE

Structure → Name

Molecular formula (counted): **C7H14**

(3 + 4 = 7 C, one ring: C_nH_{2n})

1 · NUMBERED PARENT



(Parent highlighted; carbons numbered from the end that gives the lowest locants)

2 · NAMING TRAIL

Parent: 3 C → cyclopropane (ring). *(The ring is the parent in this activity, even though the open chain has more carbons than the ring. Other paths of the same length: none.)*

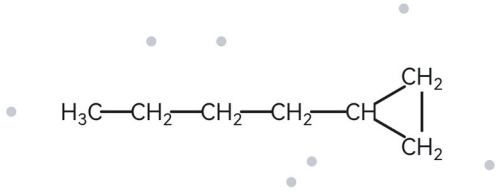
Numbering: no locant is needed. *(A single substituent on a ring is automatically assigned position C1.)*

Substituents (A→Z): butyl on C1 (the 4-carbon chain).

3 · MY NAME (TYPED BEFORE OPENING THE TOOL)

butylcyclopropane

4 · THE ALKANE CHECK (RETURNED NAME)



butylcyclopropane

Tool: The Alkane

Returned name: butylcyclopropane

Verdict: [✓] Match [ ] Mismatch

[✓] longest chain/ring confirmed [✓] lowest locants [✓] alphabetical order [✓] punctuation [✓] formula counted

B01 FOUNDATION

Name → Structure

Given name:

2-methylpropane

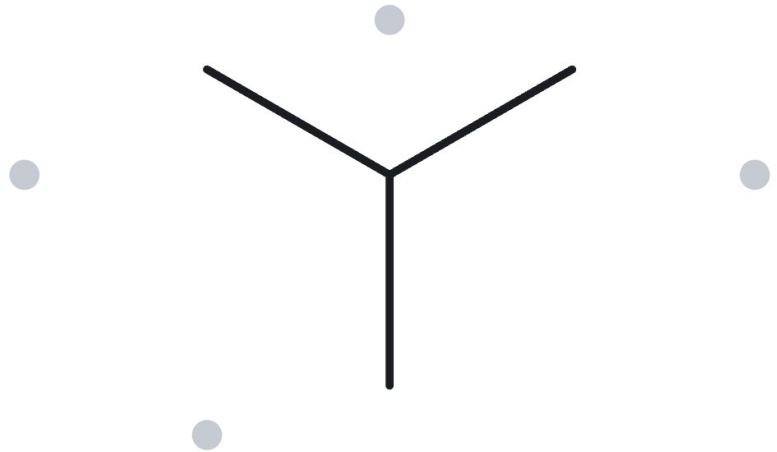
1 · NAME DECODED

Parent: root = propane → 3 C; chain

Substituents: methyl on C2

Expected formula: total C = 4; C4H10

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



2-methylpropane

3 · FORMULA CHECK

Counted from drawing: C4H10

General formula: chain, n = 4: C_nH_{2n+2} = C4H10

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: 2-methylpropane *(same parent, branches and positions as the given name)*

Verdict: [✓] Same as given [ ] Different

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded

B02 FOUNDATION

Name → Structure

Given name:

3-ethylpentane

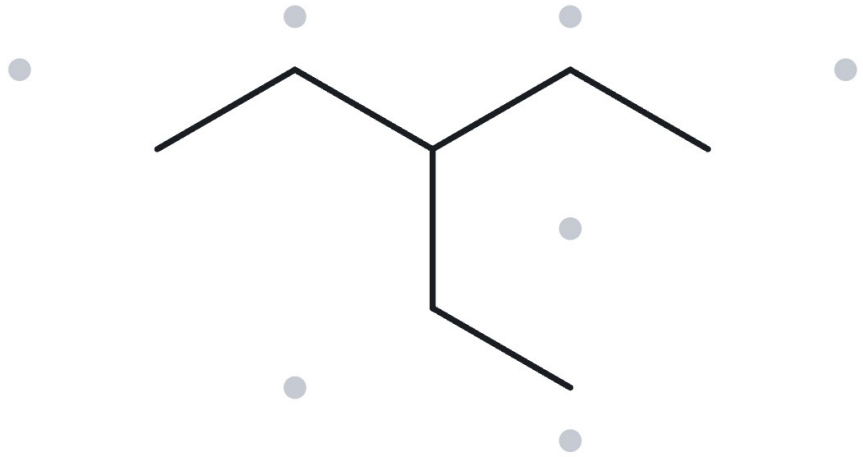
1 · NAME DECODED

Parent: root = pentane → 5 C; chain

Substituents: ethyl on C3

Expected formula: total C = 7; C7H16

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C7H16

General formula: chain, n = 7: C_nH_{2n+2} = C7H16

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: 3-ethylpentane *(same parent, branches and positions as the given name)*

Verdict: [✓] Same as given [ ] Different

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded

B03 FOUNDATION

Name → Structure

Given name:

2,2-dimethylbutane

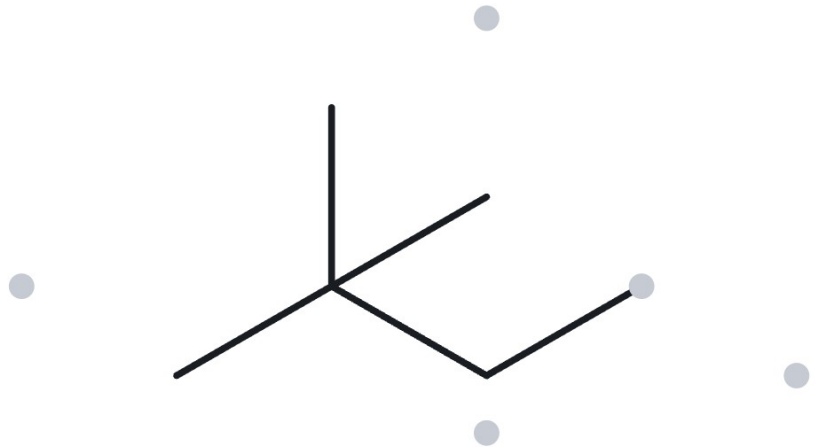
1 · NAME DECODED

Parent: root = butane → 4 C; chain

Substituents: methyl on C2, methyl on C2

Expected formula: total C = 6; C6H14

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C6H14

General formula: chain, n = 6: C_nH_{2n+2} = C6H14

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: 2,2-dimethylbutane *(same parent, branches and positions as the given name)*

Verdict: [✓] Same as given [ ] Different

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded

B04 DEVELOPING

Name → Structure

Given name:

3-ethyl-2-methylpentane

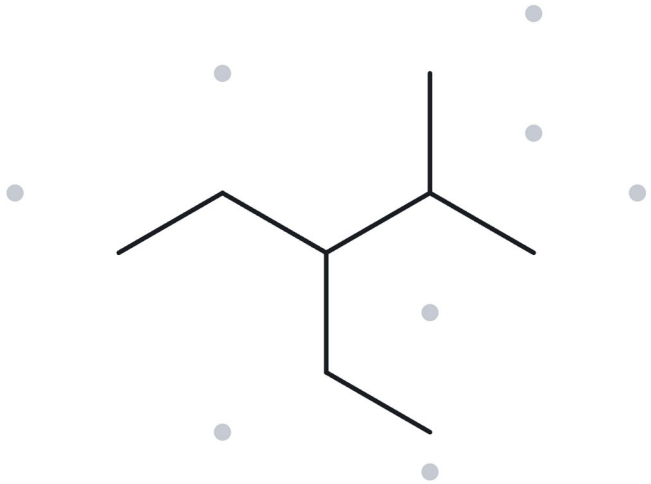
1 · NAME DECODED

Parent: root = pentane → 5 C; chain

Substituents: methyl on C2, ethyl on C3

Expected formula: total C = 8; C8H18

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



3-ethyl-2-methylpentane

3 · FORMULA CHECK

Counted from drawing: C8H18

General formula: chain, n = 8: C_nH_{2n+2} = C8H18

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: 3-ethyl-2-methylpentane *(same parent, branches and positions as the given name)*

Verdict: [✓] Same as given [ ] Different

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded

B05 DEVELOPING

Name → Structure

Given name:

1-chloro-2-methylpropane

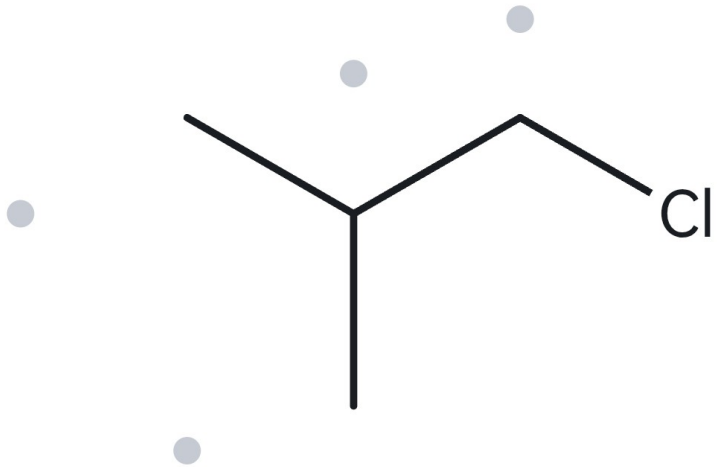
1 · NAME DECODED

Parent: root = propane → 3 C; chain

Substituents: chloro on C1, methyl on C2

Expected formula: total C = 4; C4H9Cl *(C4H10 with one H replaced by Cl.)*

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



1-chloro-2-methylpropane

3 · FORMULA CHECK

Counted from drawing: C4H9Cl

General formula: chain, n = 4: C4H10; one H replaced by Cl = C4H9Cl

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: 1-chloro-2-methylpropane *(same parent, branches and positions as the given name)*

Verdict: [✓] Same as given [ ] Different

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded

B06 DEVELOPING

Name → Structure

Given name:

1,2-dimethylcyclopentane

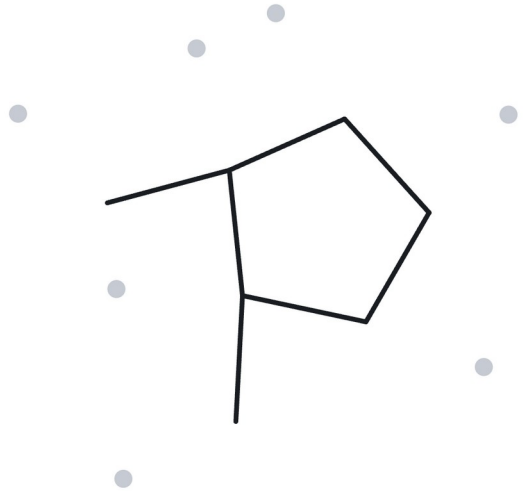
1 · NAME DECODED

Parent: root = cyclopentane → 5 C; ring

Substituents: methyl on C1, methyl on C2

Expected formula: total C = 7; C7H14 *(The two methyls are on neighbouring ring carbons.)*

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



1,2-dimethylcyclopentane

3 · FORMULA CHECK

Counted from drawing: C7H14

General formula: one ring, n = 7: C_nH_{2n} = C7H14

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: 1,2-dimethylcyclopentane *(same parent, branches and positions as the given name)*

Verdict: [✓] Same as given [ ] Different

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded

B07 DEVELOPING

Name → Structure

Given name:

4-ethyl-3,3-dimethylheptane

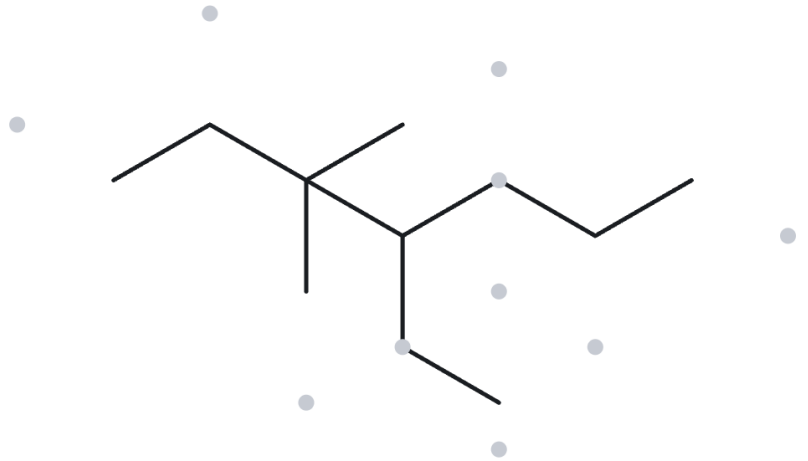
1 · NAME DECODED

Parent: root = heptane → 7 C; chain

Substituents: methyl on C3, methyl on C3, ethyl on C4

Expected formula: total C = 11; C11H24

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



4-ethyl-3,3-dimethylheptane

3 · FORMULA CHECK

Counted from drawing: C11H24

General formula: chain, n = 11: C_nH_{2n+2} = C11H24

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane *(same parent, branches and positions as the given name)*

Verdict: [✓] Same as given [ ] Different

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded

B08 CHALLENGE

Name → Structure

Given name:

tert-butylcyclohexane

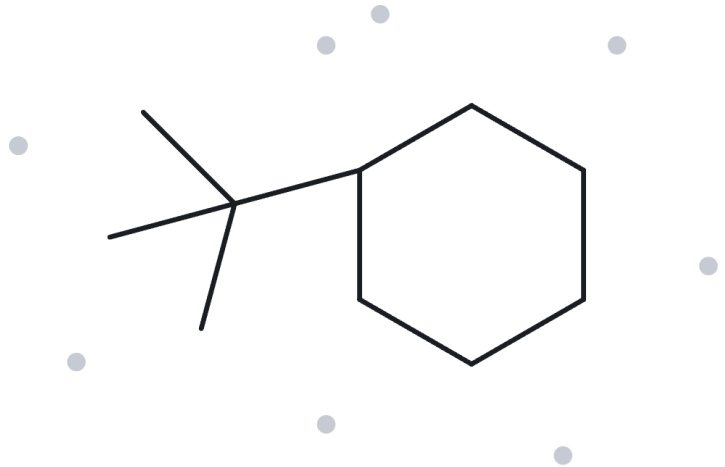
1 · NAME DECODED

Parent: root = cyclohexane → 6 C; ring

Substituents: tert-butyl on C1

Expected formula: total C = 10; C10H20 *(tert-butyl is a 4-carbon group: a carbon bearing three methyls. Systematic forms: (1,1-dimethylethyl) or 2-methylpropan-2-yl.)*

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



(1,1-dimethylethyl)cyclohexane

3 · FORMULA CHECK

Counted from drawing: C10H20

General formula: one ring, n = 10: C_nH_{2n} = C10H20

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: (1,1-dimethylethyl)cyclohexane *(same compound: tert-butyl is written (1,1-dimethylethyl), an accepted equivalent form, Appendix B.3)*

Verdict: [ ] Same as given [✓] Different wording, same compound

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded

B09 CHALLENGE

Name → Structure

Given name:

3-ethyl-1,1-dimethylcyclohexane

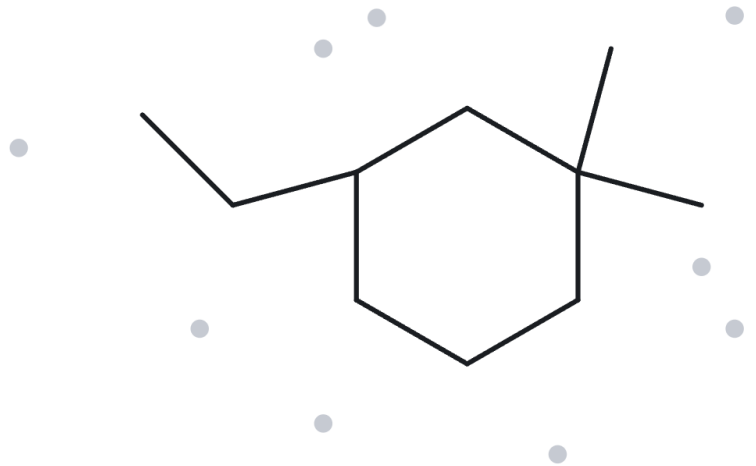
1 · NAME DECODED

Parent: root = cyclohexane → 6 C; ring

Substituents: methyl on C1, methyl on C1, ethyl on C3

Expected formula: total C = 10; C10H20 *(Both methyls are on the same ring carbon, C1.)*

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



3-ethyl-1,1-dimethylcyclohexane

3 · FORMULA CHECK

Counted from drawing: C10H20

General formula: one ring, n = 10: C_nH_{2n} = C10H20

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane *(same parent, branches and positions as the given name)*

Verdict: [✓] Same as given [ ] Different

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded

B10 CHALLENGE

Name → Structure (deliberately incorrect name)

Given name:

2-ethylbutane

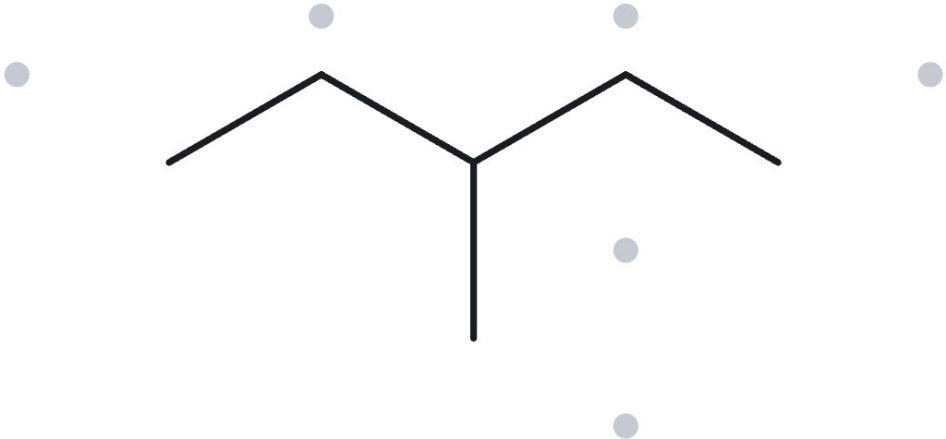
1 · NAME DECODED

Parent: root = butane → 4 C; chain

Substituents: ethyl on C2

Expected formula: total C = 6; C6H14 *(Deliberately incorrect name: drawing it as written reveals that the true longest parent chain is 5 C.)*

2 · BOND-LINE STRUCTURE (MY DRAWING IN THE ALKANE)



3-methylpentane

3 · FORMULA CHECK

Counted from drawing: C6H14

General formula: chain, n = 6: C_nH_{2n+2} = C6H14

Agrees with expected formula: [✓] Yes

4 · BACK-CHECK (THE ALKANE)

Returned by The Alkane: 3-methylpentane (the same compound as my drawing).

Verdict: [ ] Same as given [✓] Different

Correct name: 3-methylpentane.

Rule broken: the parent must be the longest continuous chain (5 C, not 4 C).

[✓] parent length matches [✓] substituent positions match [✓] formula agrees [✓] back-check recorded